

# Economic Theories of Marshall and Keynes

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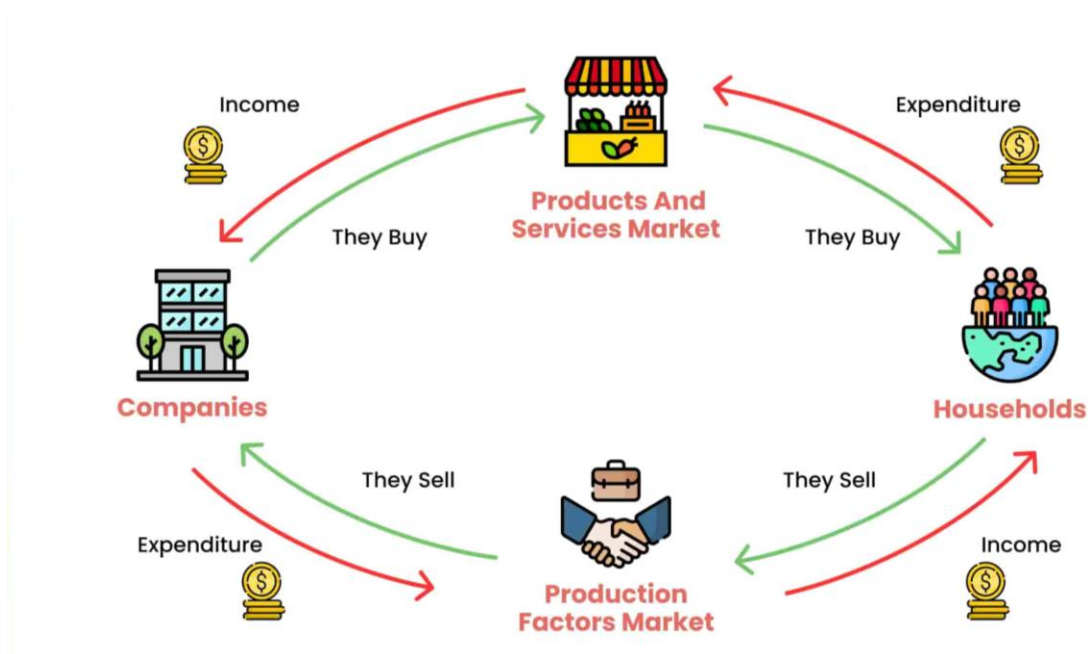
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## Alfred Marshall (1842-1924)

Workplace: Teacher of Economics at Cambridge University

Book: Principles of Economics (1890)

### Revisiting the Simple Circular Flow Diagram



The utility theory, although succeeded in incorporating the demand side within the framework of commodity's value determination, remained deficient in terms of explaining how rational, maximizing individual decisions led to the development of a self-adjusting, efficient economic process. In order to understand the development of such analysis, it is imperative to revisit the quite known "Circular Flow Diagram".

The simple circular flow diagram can be constructed using two entities, namely –households and firms, excluding the government, the banking sector and the foreign sector of economy. The diagram depicts two continuous circular flow between these entities, which are –

1. Real Flow
2. Monetary Flow

Real flow demonstrates the conversion of factors, owned by households, into final commodities, consumed by households. Therefore, it pertains to real goods and services. Monetary flow,

contrarily, shows the circulation of returns. Households are paid by firms as per the former's productivity of respective factors, which constitute the economy's income share. Firms, on the other hand, receive payments from households in exchange of the final commodities produced. An aggregation of such payments renders the total expenditure of a closed economy operating under Laissez-Faire. Since every unit of currency earned is used up to purchase commodities, income of households equates their expenditure.

The households, in order to purchase commodity, must earn and thereby sale their productive factors. For instance, a labor must work (i.e. exert physically or mentally) to earn income, and give up leisure. With increased income, they increase their utility via commodity purchase, and by giving up leisure, they face disutility. It is a worker's "rational" decision to "maximize" the excess of utility received from income over the disutility incurred by working.

Firms, on the other hand, aims to earn as much revenue as possible by selling their commodities, while trying to reduce their cost of production. Hence, a firm's "rational" decision is to "maximize" this difference between its revenue and cost.

Hence, households become the focal point of "real flow", maximizing utility; whereas firms remain the focal point of "monetary flow", maximizing profit. The mathematical principle of maximization (or optimization, more generally) allows a similar neoclassical analysis for both households and firms. Such an analysis, conducted by Marshall, completed the self-adjusting mechanism of the Neoclassical Economics.

## Unique Contribution: Extending Utility Theory to Demand Theory

Marshall's contributions to utility theory to those areas in which he went beyond Jevons, Menger, and Walras, like the latter three theorists, he formulated the notion of diminishing marginal utility as well as the necessary conditions for consumer utility maximization through exchange. Neither Jevons nor Menger succeeded in directly linking their utility theory to demand theory. While Walras had shown such a link, Marshall was much more successful than Walras in rigorously deriving the contemporary neoclassical conclusions of demand theory from his notion of diminishing marginal utility.

Marshall deduced a negative demand curve in the following manner: first, as with all utility theorists, he assumed that during the period being analyzed the individual's schedule of marginal utilities remained unchanged. Second, he assumed that an individual's marginal utility of money was given and constant throughout the period of analysis. By assuming a fixed

marginal utility of money, Marshall found a connecting link between utility schedules and price schedules.

For example, if a person has a marginal utility of two "utils" for \$ 1 of money, and one additional loaf of bread would yield four utils while one additional pound of steak would yield six utils to that person, then he or she would be willing to give up \$2 (i.e., four utils) for a loaf of bread and \$3 (i.e., six utils) for a pound of steak. The utility of a second additional loaf of bread might fall to two utils and that of a second pound of steak to three. In that case, he or she would be willing to pay \$ 1 (two utils) for the second additional loaf of bread and \$ 1 .50 (three utils) for the second additional pound of steak. This was because "the marginal utility of money to him is a fixed quantity, so that the prices he is just willing to pay for two commodities are to one another in the same ratio as the utility of those two commodities.

#### His PoV on the utility maximization procedure

Marshall extended the idea of Jevon, Menger and Walras in developing is utility maximization procedure. According to him, one half of the utility maximization is concerned with the consumption of goods and services, the process which are mentioned above. And the other half of utility maximization problem of household is concerned with the sale of factor of production.

- According to Marshall, among the three factors of production that household possesses, land-labor-capital, only the capital and labor comes with certain degree of disutility or negative utility.
- Labor: When labor was sold, there was always a point beyond which "the marginal disutility of labor generally increases with every increase in its amount. This increasing disutility of labor might come from the bodily pain and mental fatigue that a labor endures during providing labor service.
- Capital: Investing a capital in any production process requires abstinence from current consumption which also produces disutility.

The argument of Marshall regarding the disutility which arises from capitalists' abstinence is somewhat similar with the argument of Senior. Except a small adjustment.

## From Abstinence to Waiting – A Time Dimension

Marshall wanted to replace the word “Abstinence” with “Waiting” while providing his theory regarding the capitalists’ investment. As he quoted;

*“The sacrifice of present pleasure for the sake of the future, has been called abstinence by economists . . . since, however, the term is liable to be misunderstood, we may with advantage avoid its use, and say that the accumulation of wealth is generally the result of a postponement of enjoyment, or a waiting for it”*

He found the word “waiting” more appropriate as it better describes the circumstances of capitalists, who refrain themselves from current consumption, hoping or waiting for a much better consumption in the future which will provide better utility to them.

Thus, Marshall concluded that, people continuously calculate the appropriate exchange involved in minimizing the pain of labor service or waiting against the pleasure received from current consumption. Laborers decide how much painful exertion to endure and capitalists decide how much painful waiting to endure, based on their respective marginal utility.

## Symmetry Between the Household and the Firm (Neoclassical Theory)

### Substitutability as a Tool in Utility Maximization

The equimarginal principle of utility theory dictates that the marginal utility derived from each dollar spent on every commodity must be equal. Mathematically,

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2} = \dots = \frac{MU_n}{P_n}$$

Such an equality can only be achieved because of substitutability. If marginal utility per dollar derived for commodity 1 remains higher than that of commodity 2, then the consumer will reallocate resources to increase his consumption of commodity 1, the good that renders higher utility, upon reducing his consumption of commodity 2. The increased consumption of good 1 diminishes its marginal utility, while that of good 2 increases due to its reduced consumption. Such redistribution in consumption pattern will prevail until the equality is achieved, and utility is maximized.

This entire analysis is based on the assumption that consumable commodities are **substitutable** in terms of utility. The assumption is rightly made, since within the current economic structure a

consumer faces wide arrays of choices among goods. Therefore, when price of one good increases, it is possible for the consumer to replace it with some other close, even if imperfect, substitute that can increase their utility.

### Classical Assumption of Fixed Technical Coefficients in Production

Most classical economists, including Menger and initially Walras, assumed factors of production to not be substitutable. In Ricardo's Theory of Rent, it was asserted that lack of fertility in land was substituted by additional labor and capital to maintain the same level of production. Despite that, by and large economists believed that factors had to be used in fixed proportion. This meant that with a given capital-labor ratio (suppose, 1:1), a decrease in wage will not allow producers to hire more labor unless they increased their capital in the same ratio. Likewise, if wage increased, producers did not have the freedom to substitute labor's productive capacity with that of capital. This lack of substitutability made it impossible for the classicals to compare a firm's profit maximizing decision with an individual's utility maximization decision.

### Marshall's Integration of Utilitarian Economic Theory in the Context of Production

Marshall founded the case of firm's profit maximization on the following arguments:

1. Substitutability of productive factors
2. Law of diminishing returns

Marshall viewed firm's production decision in a new light, juxtaposing it with the utility theory, and established that factors of production were substitutable just the way consumer goods were. This allowed him to take into account firms' flexibility in production decision, and analyze how they maximized their profits. Additionally, he inferred that factors, with more employment, exhibited diminishing returns. This too, was aligned with the theory of diminishing marginal utility. And in that process, he established the symmetry between the maximization analyses of households and firms.

Subsequently, a framework of utility maximization analysis and that of cost minimization are juxtaposed to demonstrate their similarity.

As it can be seen, indifference curve alongside isoquant is downward sloping and convex to the origin, which reflects Marshall's argument on substitutability and diminishing returns respectively. The framework that allows the analysis of utility maximization with respect to a budget constraint, allows that of quantity maximization with respect to the production cost.



## Firm's Maximization Problem – Two Vantage Points

A firm's maximization problem can be viewed from either of the two vantage points, namely – its outputs or its inputs.

Scrutinizing it from the output point of view, one has to study the total revenue and total cost entailed in its production. In that process, scrutiny of average and marginal revenues and costs must be conducted as well. Marshall developed his theory of firm from this vantage point.

The second point of view required an analysis of value produced by the hired factors, along with the cost of hiring those factors. This input vantage point operated as the basis for "marginal productivity theory of distribution", which reflected on the neoclassical implications for distribution of income. P.H. Wicksteed and John Bates Clark independently developed this theory.

## Marshall's Theory of Firm

Marshall's theory of firm is very crucial for determining the price in his analysis. He provides few assumptions before proceedings;

1. Prices were determined by supply and demand
2. Demand was determined by consumer's utility
3. Supply was determined by firm's cost schedule
4. The firm is a competitive firm, it takes prices as given.

Marshall uses the concept of a "representative firm" in his analysis. A representative firm is an average firm in the market, which had no special advantages or disadvantages, hence its cost of production reflects the average cost of production of every firm in the industry.

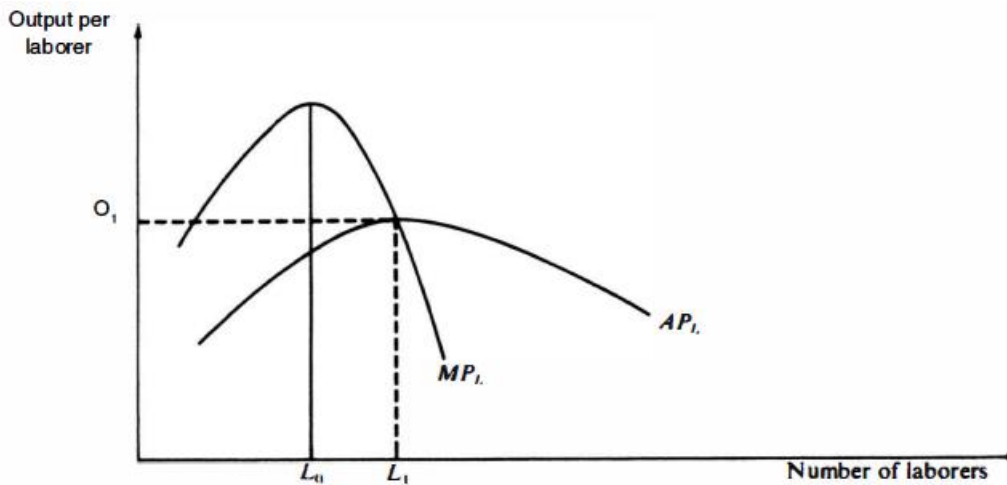
His analysis was based on a distinction between three time periods;

1. Market Period: Where Supply is fixed, and the price is solely determined by the demand of that particular commodity
2. Short period (Short Run): Capital was fixed, but the supply could be changed by altering the number or labor employed
3. Long Period (long Run): Supply could be changed by altering both the labor and capital

The market period was relatively unimportant in Marshall's analysis, so we will confine our discussion to the behaviour of the firm in the short and long periods. Throughout the discussion, it was assumed that the firm had so many competitors that it could not itself directly influence either the prices that it had to pay for inputs or the price that it received for its output

### Firm's Production and Cost Curves in the SR

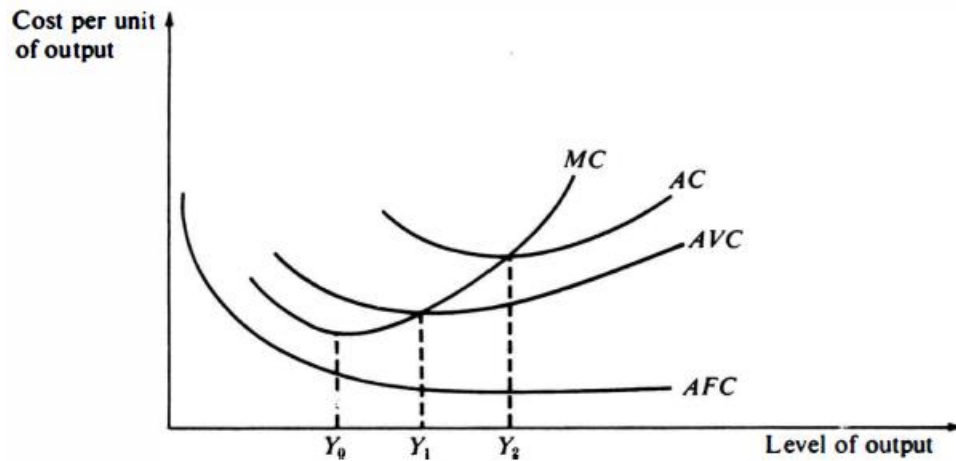
The following figure depict the relationship between a firm's marginal product and average product with respect to the number of labors hired.



The marginal Productivity of labor continue to rise up to the point  $L_0$ , because of the relatively large capital labor ratio. Beyond  $L_0$ , marginal productivity decreases but average productivity continues to rise up to the point  $L_1$  as the marginal product is higher than the average.

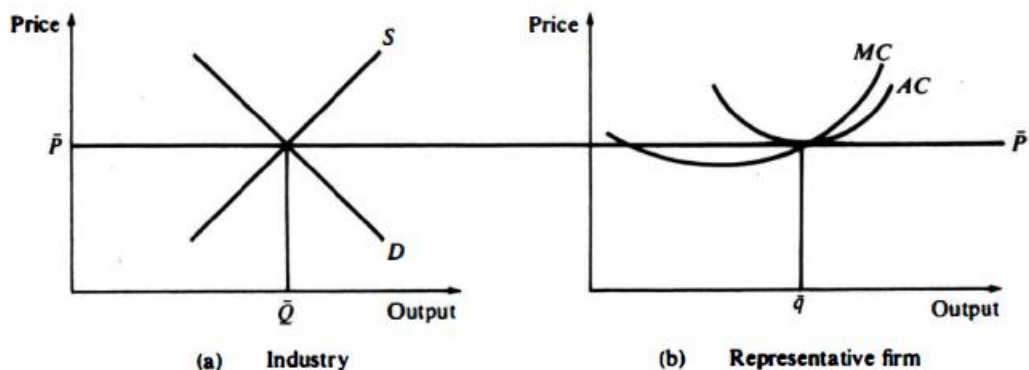
Following figure depicts the cost curve of a representative firm;

As the figure illustrate, the Average variable cost (AVC) is minimized at production level  $Y_1$ , but the Average Cost (AC) minimizes at point  $Y_2$ . That is because beyond point  $Y_1$  to the point  $Y_2$ , the increase in variable cost is more than compensated by the decrease in fixed cost. The marginal cost curve intersects both the average cost and the average variable cost curves at their minimum point.



### Equilibrium in the Short Run

As from the discussion of Marshall, the profit maximizing condition for a competitive firm in the short run is,  $P = MC$ .



The left panel in above figure shows the price determination in the industry with the forces of supply and demand. In the right panel, the representative firm takes that price as given, and determine its profit maximizing output where the price line coincide with its MC and AC.

In the short run, the average cost of a representative firms is equal to its price, that does not mean that the firm do not earn any profit in equilibrium. Because neoclassical economists insist that a normal, or average, rate of profit on capital is a necessary part of the firm's capital costs, then such a profit would be included in average costs (as a part of its average fixed costs component). It is important to keep in mind that, the output of a representative firm is much smaller compared to the output of the whole industry.

**Quasi-Rent:** If a firm earn more than normal profit, or its average cost curve is below the price, due to some natural advantages, then those higher profit will be called as quasi rent. Marshall argued quasi rent to be similar with the “Ricardian Rent” which were earned by the owner of the superior land.

### Equilibrium in the Long Run and the problem of Competition

According to Marshall, all factors of production can be altered in the long period or in the long run. Difference between Short Run and Long Run is that – firms’ size can be adjusted to achieve productive efficiency.

Hence Marshall argued that, due to competition and invisible hand;

1. Rate of profit is equalized in all firms; cost of production is minimized (by achieving productive efficiency)
2. Consumers can buy product at minimum price, a price which just covered the socially necessary cost of production yielded no surplus for any class to expropriate from other class

Thus, Marshall argued that there is “universal advantage” and social harmony inherent in the process of competitive exchange.

### Returns to Scale: Problem in the long run

Hunt argued in his writings that, the commonsensical conclusion of the neoclassical ideologies should be a constant return to scale (Proportionate Change in output due to proportional increase in all input).

However, Marshall has explained in his idea, how decreasing return to scale and increasing return to scale could occur in any competitive settings.

**Decreasing Return to Scale (Less than Proportionate Change in output due to proportional increase in all input):** where land or natural resources were important in a production process, there was a tendency toward decreasing returns to scale (or increasing average costs over the long period).

**Increasing Return to Scale (More than Proportionate Change in output due to proportional increase in all input):** when capital and labor were the primary factors of production, there was a tendency toward increasing returns to scale (or decreasing average costs over the long period).

Marshall showed that, the increasing efficiency of greater size come from two sources;

- **Internal Economies:** Where the increase of the size of the firm may lead to improved organizational structure, greater opportunity to specialize the factor of production and make greater division of labor.
- **External Economies:** Benefit received by a firm, due to the pricing and locational decision made by other firms, Knowledge can spill over across firm, and the benefit of closely interrelated secondary and tertiary industries (Forward and Backward Linkage).

## Marshall's Contradiction Pointed out by Hunt

Hunt argued that, the tendency of increasing the size of firm to achieve productive efficiency will lead to the establishment of monopoly and oligopoly. So, Marshall's theory should lead him to one of the three possible conclusions;

1. Marshall could have abandoned the general utilitarian argument, that the invisible hand maximized the overall social utility, and then defended capitalism on the basis of a new ideology stressing the social advantages of giant oligopolistic business corporations.
2. Marshall could have argued that the social advantage of competition was more significant than the social advantage of productive efficiency and hence advocate government intervention to prevent large corporation,
3. Marshall could have taken Marx's view and promote socialism, by which government will be the owner of the large corporation, and thus ripping the benefit of large-scale production without establishing monopoly or oligopoly.

As hunt stated, Marshall did not accept any of those conclusions. He more or less took a balanced path like Mill, where he did not bash the poor like Malthus or Senior, rather he incorporated some principle of moral and social philosophy to address the hardships of poor, which was according to hunt, incompatible with his utilitarian theory.

## Marshall's Ideological Defense of Capitalism

- Marshall's defenses were based on Evolutionary Social Darwinism where he showed the mechanism of a slow societal progress
- He argued that, any attempt to change the society will miserably fail
- He showed Similarities between caste system and capitalism (appeared exploitative but progressive in feature) as well as the Evolutionary theory of life cycles of natural organisms (young tree)

- He also Defended the doctrine of the Invisible Hand by arguing that workers can become capitalist by having appropriate moral virtue.
- According to him, Capitalism was crueller in the past, but now it becoming more concerned for the poor day by day.

## John Maynard Keynes

John Maynard Keynes Provided his thoughts and insights on capitalism when the American Economy was in turmoil and the whole structure of capitalism, utilitarianism and neoclassical ideas were on a verge of collapsing. At that time people started to think that the capitalist economy is a gone case, and the theories provided by Marx and other socialist activist were going to be proved superior.

Before diving into the Keynes theoretical structure fully, some of the key neoclassical features, their similarities and dissimilarities with Keynes will be discussed.

## Three Principal Ideology of Neoclassical Utilitarianism

Neoclassical ideas always defended the Laissez-faire capitalist idea. The three basic ideologies behind that were:

1. Marginal Productivity Theory of Distribution, which pictured competitive capitalism as an ideal of distributive justice
2. "Invisible Hand" which pictured competitive capitalism as an ideal of rationality and efficiency
3. The faith in automatic, self-adjusting nature of market, where the governments' role was to enforce contracts and to defend the powers and privileges of private properties

The first two tenets were pure blessing for capitalist. But the third one was a mixed blessing. The question is now why? In the following analysis, we will try to cover that discussion

- During the early stages of Capitalist Industrialization, laissez-fair was desired by the most of the capitalists, because:
  - i. Government mostly represented the interest of older merchants and landowners, because they had more influences on the government.
  - ii. Under the banner of laissez-fair, capitalist could promote the uninterrupted profit-making activities and make those campaigns look like activities to promote public's freedom from tyrannical government abuse.
- During late 19<sup>th</sup> century, development of world-wide capital market and improvement in production and transportation led to immense concentration of industrial power in giant corporations and cartels. That had two important consequences:
  - i. Unregulated competition became extremely costly and hazardous to these giant corporations,
  - ii. Due to which, giant corporations significantly reduced whatever amount of flexibility and adjustability the market had previously possessed.

### Transition in the Power Dynamics and the urge for regulation

Now it can be established that, at that time, the belief in self-regulatory system were decreasing and it was becoming increasingly costly for the other capitalist. At the same timeline, one significant transition took place in the power dynamics.

- During late 19<sup>th</sup> and early 20<sup>th</sup> centuries, the government were being controlled by capitalist.
- So, to get rid of the market anarchy and the unhealthy self-regulating structure of the market, they turn to the government to come up with regulations and interventions.

Interstate Commerce Act (1887), Sherman Antitrust Act (1890) were examples of such interventions. Although, up until that point, those interventions were merely intended to ensure smooth and efficient functioning of “Invisible Hand” termed as the situation “Workable Competition”.

However, instability of capitalism grew worse as time progress. Following table depict the number of recession that US and UK had during first and second half of the nineteenth century.

| Timeline                              | Country | No. of Recession | Year |
|---------------------------------------|---------|------------------|------|
| During the first half of 19th Century | US      | 2                | 1819 |
|                                       |         |                  | 1837 |
|                                       | UK      | 4                | 1815 |
|                                       |         |                  | 1825 |
|                                       |         |                  | 1836 |
|                                       |         |                  | 1847 |

| Timeline                               | Country | No. of Recession | Year |
|----------------------------------------|---------|------------------|------|
| During the Second half of 19th Century | US      | 5                | 1854 |
|                                        |         |                  | 1857 |
|                                        |         |                  | 1873 |
|                                        |         |                  | 1884 |
|                                        |         |                  | 1893 |
|                                        | UK      | 6                | 1857 |
|                                        |         |                  | 1866 |
|                                        |         |                  | 1873 |
|                                        |         |                  | 1882 |
|                                        |         |                  | 1890 |
|                                        |         |                  | 1900 |

In the 20<sup>th</sup> Century, the situation grew worst and eventually led to the great depression of 1930.



## The Great Depression of 1930

Due to the massive decrease of the stock prices at the New York Stock Exchange, the American economy was in a turmoil. Businesses lost their confidence, cut back their production, which created massive unemployment which brought catastrophe to the global economy and fell into a crisis which lasted for nearly a decade.

### Event during the period 1929-1932

- 85000 business failures
- 5000 banks suspended their operation
- Stock value of NYSE fell from 87 billion dollars to 19 billion dollars
- Unemployment rose to a record breaking 25%
- Manufacturing output decreased by almost 50%

What had happened to reduce the output of goods and services so drastically? Natural resources were still as plentiful as ever. The nation still had as many factories, tools, and machines. The people had the same skills and wanted to put them to work. Yet millions of workers and their families begged, borrowed, stole, and lined up for a pittance from charity, while thousands of factories stood idle or operated at far below capacity. The explanation lay within the institutions of the capitalist market system. Factories could have been opened and men put to work, but they were not because it was not profitable for businessmen to do this. In a capitalist economy, production decisions are based primarily on profits not on people's needs.

In that situation, it became clear to the many of the social thinkers that the myth of self-adjusting market is not functioning anymore. Moreover, it was becoming a threat to capitalism itself. Everyone was feeling that, a drastic measure was necessary, but the neoclassical ideologies did not have any solve of that grave problem, according to the classical theory, depression did not even occur! So how a remedy would come from them.

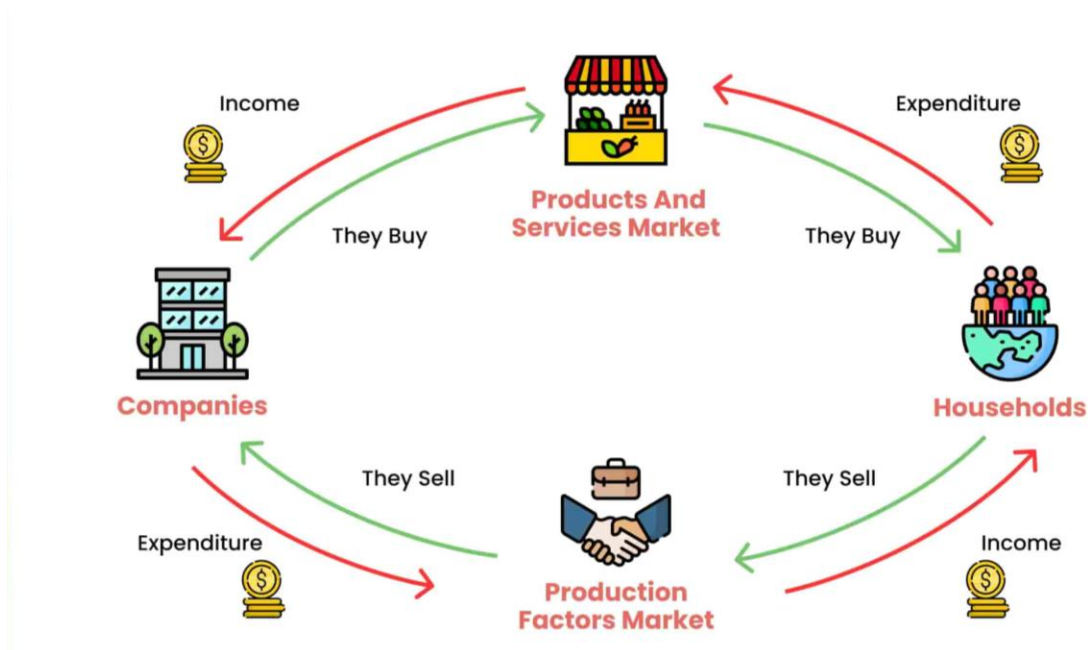
## Arrivals of Keynes and His Remarks on Marx

In the year 1936, Keynes published his famous book, *The general Theory of Employment, Interest and Money*. In his book, he tried to show the flows on current capitalist system and the adjustments which were needed to preserve the system.

In his way to understand the depression, Keynes provided some remarks about Karl Marx. According to Keynes, the ideologies of Marx are fraudulent and logical fallacy. He had no interest in Marx's discussion as a scientific theory. He even quoted that, "how a doctrine so illogical and so dull can have exercised so powerful and enduring an influence over the minds of men, and through them, the events of history" However, he preferred Malthus over Ricardo.

## Theoretical Setting of Keynes Analysis

The conceptual context of Keynes' theory was as same as Walrasian General Equilibrium theory. The analysis of a continuous process of production, circulation and consumption, the process which is showed in a circular flow diagram. In a given year or period, an individual firm produces some amount of good. By selling them, the firm pays its cost of production (wages, rent, salaries etc.). What remains after cost is profit.



A cost of production to the firms, represent income of an individual or another firm. The profit is also an income, the income of the business owner. So, according to the circular flow of the economy, the value of what has been produced must be equal to the income generated to produce it. For the entire economy, the scenario should be, the value of everything produced in an economy for a given period, should be equal to the total of all income received in that period. So, if businesses in an economy wants to sell all of their production for a given period, then people must spend all of their income in an aggregate.

Money flows from business to public in the form of wages, salaries, rents, interest and profit. This money then flows back to business in terms of payment for goods and services. However, this circular flow has three leakages:

1. Saving: People do not always spend all of their income, some of the portions are saved.
2. Import: People may spend money to buy foreign products.
3. Taxes: The taxes that people pay are withdrawn from the flow

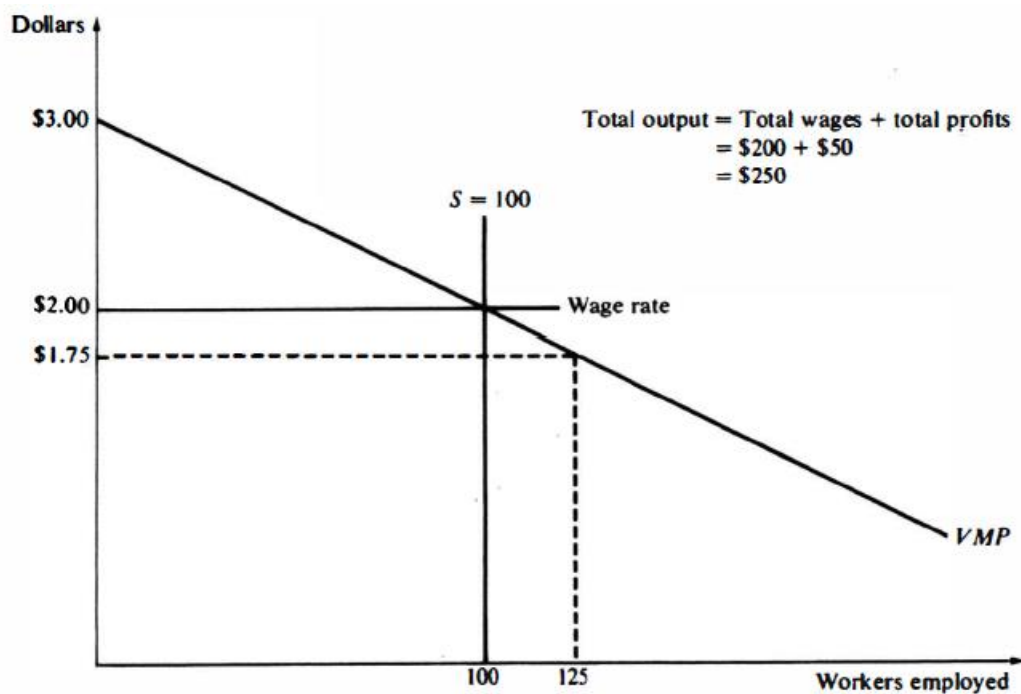
These three leakages may be offset by three spending injection,

1. Export: When foreigner buy domestic good
2. Investment: Businessman will borrow the saving of the public and use that to fund their capital expansion
3. Government Expenditure: Fund raised from taxation will be used to finance government spending

This was the orthodox neoclassical version of how competitive capitalism normally functioned.

## Neoclassical view of Wage Rate and Total Output

The following graph represent the determination of wage rate and total employment, hence the total output of an economy according to the neoclassical ideology. In the figure, the downward slopping curve, VMP, represent the marginal value product of the labor. The wage rate is determined by the intersection point of VMP and the labor supply curve.



If the labor supply is 100, then the wage rate would be 2 dollars. Now if 125 workers want to find job, then the wage rate will decline to 1.75 dollars. Thus, given a demand curve for labor, the wage of the labors is determined by the labor supply. If more labor wishes to find job, they have to accept lower salary. That was the idea possessed by the neoclassical economist.

According to them, if unemployment existed in an economy, that was because the worker refused to work, unless they receive more wage than the value of their marginal product. So, according to neoclassical economist, if all of the worker accepts wage equal to their value of marginal product, profit maximizing behavior of the firms ensures that, there will not be any involuntary unemployment.

If workers were willing to accept a wage that was equal to the value of their marginal product, then there would never normally be any problem of aggregate demand being equal to aggregate supply. All income would normally be spent. The three injections into the income-expenditure flow would normally equal the three leakages. This was because:

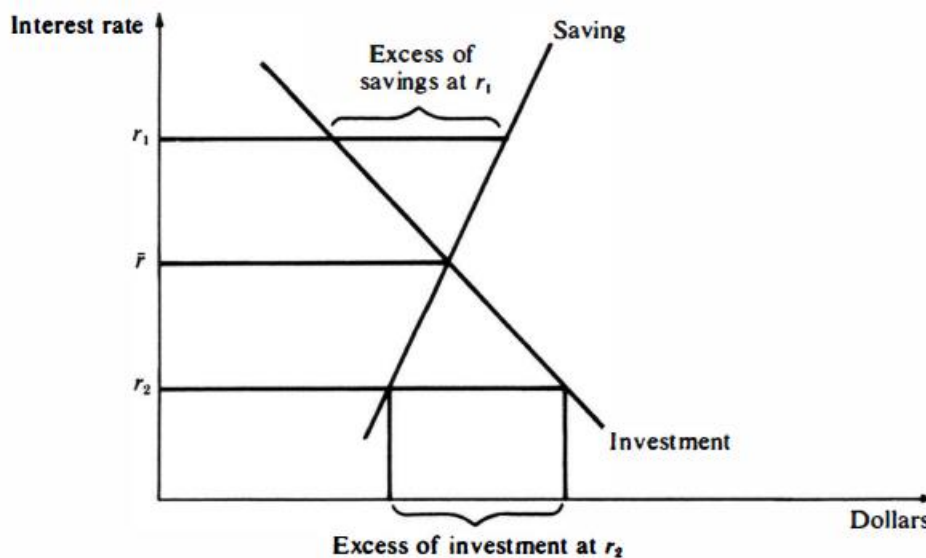
1. Neoclassical theory showed that the free play of supply and demand would balance international transactions
2. "Sound fiscal policy," in the view of both neoclassical economists and most politicians, dictated that the government maintain a balanced budget in which taxes equaled government expenditures
3. The rate of interest would always bring saving and investment into equality.

However, Keynes and the neoclassical economist possesses significant differences regarding the third point.

## Neoclassical View of Interest rate, Saving and Investment

According to neoclassical economist, people always prefer present consumption over future, so they will save money if they receive substantial reward on saving. Interest rate is that reward that induces people to save more. The higher the interest rate, the larger amount of money people will save. On the other hand, the higher the interest rate, the more money people have to pay for the fund they have invested.

As the following graph prevails, the demand supply mechanism and competitive nature of the market will always automatically create an interest rate where investment equals saving ( $S=I$ ). That means, all three leakages would automatically be brought into equality, and aggregate demand would equal aggregate supply.



The only conceivable cause of what appeared to be involuntary unemployment was, to nearly all neoclassical economists, the refusal of laborers to take sufficiently large cuts in wages. Thus, during the Great Depression, when neoclassical economists were consulted by governments regarding the most effective way to combat the depression, even those neoclassicists who were the most humane and sympathetic to the plight of workers could recommend nothing but a general cut in all wage.

### Keynes's Defense of the Marginal Productivity Theory of Distribution and Contradiction pointed out by EK Hunt

Keynes was a consistent defender of the Marginal Productivity Theory of Distribution postulated by Neoclassical Economist. The neoclassical recommendation for increasing the employment was to decrease wage rate, which will attract the firms to hire more labor in search for more profit. According to EK hunt, Keynes wanted to agree with the neoclassicists and simultaneously to disagree with them. Keynes argue that, real wage could be decreased by two ways.

1. Money Wage rate could be decreased while prices of the wage goods remained constant or decrease more slowly.
2. Price of Wage goods could be increased while the money wage remained constant or increase more slowly.

Keynes argued that worker would never accept the first method of reducing wage, but will accept the second. However, EK hunt sees the argument as unconvincing. According to hunt, workers are generally concerned with the cost of living as they are with their money wages.

Hunt also argued that, if the workers were receiving wage equal to their value of marginal product before the great depression, as Keynes believe they were, then a sharp decline in employment in 1930's should have seen a higher real wage of the employers. Because with high unemployment and fixed level of capital input, the productivity of the labor should rise sharply. But that was not the case. So, it seemed logical to hunt that, employers were receiving real wages substantially below their marginal productivity.

Hence hunt concluded that, Keynes's adherence to the marginal productivity theory that wages were always equal to the workers ' marginal productivity contradicted the rest of his theory. And according to hunt, those contradiction in theory provide best insight into the thinker's ideological orientation. Keynes wished to furnish capitalist governments with theoretical insights that would help to save capitalism. In doing so, it was necessary for him to abandon some tenets of neoclassical theory. But, as we will see, he wanted to retain as much neoclassical ideology as possible. So, he adhered to both the marginal productivity theory of distribution and the belief that the free market efficiently allocated resources (once full employment was attained), despite the fact that both these tenets of neoclassical ideology were logically tied to the belief that the free market automatically created a full employment, Pareto optimal situation. Even with theorists having the extraordinary logical ability of Keynes, ideology very frequently wins out over logic.

## Keynes's Analysis of Capitalist Depression

As noted, Keynes tried to retain as much of capitalism as possible. What did he essentially change then? Keynes discussed the underutilization of productive capacity during depressions extensively in his General Theory. He asserted that much of the physical capital remained unused, while less and less labor were employed. With that in the backdrop, he rejected the capitalist notion that starting from a situation of full employment, rate of interest would automatically equate saving and investment and thus keep aggregate demand equal to aggregate supply. His departures can be restated followingly –

1. The level of aggregate income, compared to the rate of interest, had a more significant impact on determining the amount of saving.
2. Interest rate was determined by the demand and supply of money, and not by savings and investment.

Such departures, although meant to only question the automaticity of the market forces, were severe attacks to the remaining two pillars of neoclassical ideology (marginal productivity theory of distribution and efficient allocation of resources) as well.

## Level of Aggregate Income Determining Savings

In order to prove his first departure, Keynes made use of the “consumption function” which is –

$$C = C_0 + (MPC)(Y_d)$$

Here, C depicts total consumption which is an aggregate of autonomous consumption and income-induced consumption. Income-induced consumption depends on the disposable (after-tax) income and the “Marginal Propensity to Consume” or MPC. Incidentally, MPC gives the amount of increase in consumption expenditure due to a one unit increase in disposable income. Similar concept of Marginal Propensity to Save or MPS explains how much of an additional dollar of disposable income will be saved. Mathematically,

$$MPC + MPS = 1$$

Using this interdependence between consumption and savings, Keynes showed that savings is a function of income level. In fact, he claimed that only substantial change in interest rate could affect the variable in question. In addition, neither MPC nor MPS, according to Keynes, were primarily determined by or among the primary determinants of interest rate.

## Interest Rate Determination and the Degree of Liquidity Preference

The following quote by Keynes essentially summarizes the key idea of interest rate determination. It is not the price of investment, but rather the return earned by sacrificing liquidity.

*“It should be obvious that the rate of interest cannot be a return to saving or waiting as such. For if a man hoards his savings in cash, he earns no interest, though he saves just as much as before. On the contrary, the mere definition of the rate of interest tells us in so many words that the rate of interest is the reward for parting with liquidity for a specified period.”*

Although savings and investment are closely linked to money demand and supply, they no longer found the basis for interest rate determination as per Keynes. Rather it is the liquidity preference (money demand) and the quantity of money (money supply) that equilibrates the money market.

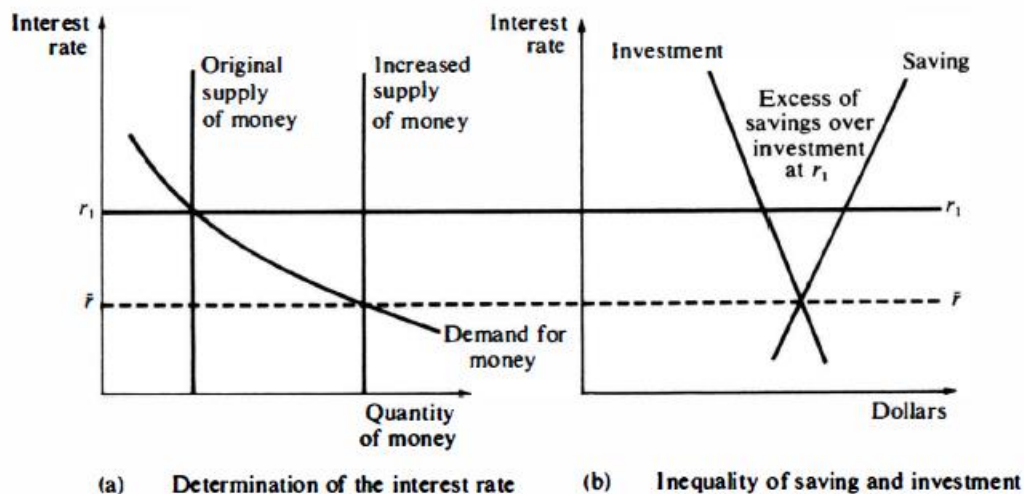
At any given time, the supply of money was constant at a level determined the monetary authorities (typically, the central bank). Keynes categorized the demand for money into three –

1. Transactions motive
2. Speculative motive
3. Precautionary motive

Transactions motive depended on the volume of transactions, which in turn depended on the level of income. Precautionary motive guaranteed the security money for future. Speculative motive, however, was an inverse function of interest rate. As interest rate rises, people become more willing to part with their liquid cash in order to earn high returns. Additionally, people can make speculations regarding the rate of interest. At low levels of interest, people expect its rise in the future, whereas at high levels of interest, people expect it to fall. Then, in the former case for instance, people hold money in liquid form with the expectation to purchase bonds once the interest rate rises. This is because existing bond prices fall when interest rate rises. Hence, with the rate about to rise, which is certain at very low levels of interest, people will certainly hold liquid money and increase speculative demand for money. Contrarily, at high interest rates, people expect it to fall in the near future and therefore buys bonds, reducing the money demand.

### Savings-investment Disequilibrium

The adjacent figure demonstrates how the interest rate, determined by money demand and supply, fails to equilibrate savings and investment.



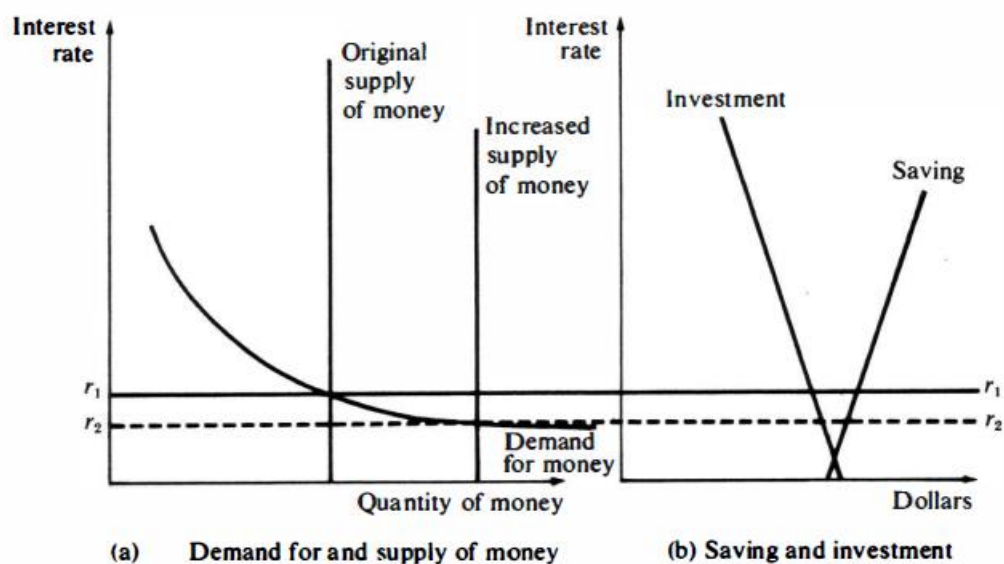
It can be seen that at equilibrium rate  $r_1$ , savings exceed investment. If this status quo is perpetuated, aggregate demand will fall short of aggregate supply which will lead to accumulation of inventories and cutbacks in production. With reduced output, unemployment increases, causing income and production level to fall further.



In that context, businesses already have excess production capacity. Hence, they have no incentive to accumulate capital and expand. This reduces investment drastically. On the other hand, with lower levels of income, savings decline more than proportionately due to much of the income being spent for meeting consumption needs. With the continuation of falling savings and investment, they will be equated at a level of low output, high unemployment and excess productive capacity.

In this situation, an increase in money supply can increase income and reduce interest rate to a point where savings equals investment but not at a low level of income. If such a policy works, one need not worry.

Keynes, however, was worried. As he had shown, in the context of depression, such monetary policies barely work. This can be the case if there exists extreme inequality in the income distribution, causing the wealthy to save a lot more than required for investment, and if the full-employment level of income remains so high that it fails to equate savings and investment despite all decrease in interest rate.



Interestingly, even without such drastic differences between savings and investment, monetary policy might be ineffective. Suppose, the rate that equates savings and investment at full employment is extremely low (below  $r_2$ ), and the prevailing interest rate is low ( $r_1$ ) as well. After taking monetary policy, rate falls and reduces to  $r_2$ . Although this rate is not sufficient to achieve the equilibrium of savings and investment, scrutinizing very low interest rate people might speculate that it can only rise in the future. With that expectation in mind, they might hold money in liquidity and not purchase bonds. Thus, the monetary instrument fails to achieve the required low level of interest. Such a case is known as "liquidity trap". In this situation, free market will fail in achieving full employment and monetary policy will appear to be ineffective.

## Solution to Depression

Marx and Hobson, much like Keynes, opined that depressions were the result of the capitalists' inability to find adequate investment opportunities that could balance the increased savings generated from increased income. Keynes contributed further in this analysis in the sense that he explained how equilibriums are created at stable but low-income levels along with high unemployment. In terms of solution, Marx did not believe there was any within the framework of capitalism, while Hobson advocated for redistribution of income to reduce savings. Interestingly, one might compare such ideas to be similar to Malthus's, although the latter advocated for more income for the landlords. The problem with Hobson's idea was that, a capitalist-dominated political system will not permit such redistribution.

Keynes too proposed the intervention of the government, but not for it to legislate and redistribute income. And neither did he ask for socialist reforms. What he asked of the government was to step in as an economic actor, borrow and spend the excess saving on socially useful projects without enhancing the economy's capital stock i.e. its productive capacity.

However, socially useful projects would not directly benefit the wealthy population as it would benefit the lower- and middle-income recipients. To simplify, such projects are indirect forms of redistribution. Therefore, it is in the interest of the government (run by the wealthy), to obstruct such projects. Keynes recognized this complication himself, and argued that it might be in the interest of the government to channel spending towards the wealthy, even though it might contradict its goals on the grounds of equity. The essence of this solution was humorously captured in Keynes's own words –

*“If the Treasury were to fill old bottles with banknotes, bury them at suitable depths in disused coal mines which are then filled up to the surface with town rubbish, and leave it to private enterprise on well-tried principles of laissez faire to dig the notes up again . . . there need be no more unemployment . . . It would indeed be more sensible to build houses and the like; but if there are political and practical difficulties in the way of this, the above would be better than nothing.”*

## Efficacy of Keynesian Policies

### The Second World War

Since the publication of General Theory in 1936, economists had been debating the relevance of Keynes's theory. Then came the second world war (WWII). During wartime, increased armament production, with all its forward and backward linkages, increased the demand for labor. Additionally, 14 million people in the American Armed Forces were to be armed, quartered and fed. Suddenly, the labor market surplus was transformed into a shortage.

This wartime experience proved the merit of Keynes's theory, as many believe. In 1946, Congress passed the Employment Act, which legally obligated the government to use its taxing, borrowing and spending powers to maintain full employment. However, after the ending of the war, the efficacy of the theory proved to be less than expected. Although, depressions commensurate to that of 1930s did not reoccur, recessions have been a common phenomenon since then. During the period of 1948-71, multiple recessions ranging from one to two years took place. Between 1973 and the early 1980s, American capitalism with most other industrialized nations suffered from stagflation. Unemployment rate ranged from 6-10 percent, while inflation ranged from 5-11 percent.

Since the end of WWII, military expenses in these advanced economies, especially the United States, have been kept high. For a period, it was to combat the Cold War, and then after its end, such expenses were to tackle "rogue states" and "terrorist network". Keynesian policies, having served quite well for a certain period of time, have created complications as well. Two of such major complications are the creations of the Military Economy and the Debt Economy.

### The Military Economy

The total expenditures in the American economy for wars and preparation for potential wars, past, present, and future, including both warfare and related expenditures, grew from \$27.9 billion per year in 1947 to \$112.3 billion in 1971. These figures represent a total of 12.2 percent of GNP in 1947 and 11.1 percent in 1971. Moreover, if one analyzes the "multiplier" effect of further aggregate demand created by these military expenditures, the impact was much larger. The most careful study of these effects resulted in two estimates, constituting lower and upper limits. The conservative estimate showed that, in 1947, military expenditures accounted, directly and indirectly (through the induced or multiplier effect), for 24.4 percent of aggregate demand in 1947 and 22.2 percent in 1971. Using the higher estimate, militarism accounted for 30.5 percent of aggregate demand in 1947 and 27.8 percent in 1971.

Why, and how has this militarism thrived in the context of American capitalism when no other form of government spending could? Militarism offers innumerable advantages to large corporations:

1. It stimulates aggregate demand without redistributing income from the rich to the poor.
2. There can never be too much elaborate weaponry; government financed research constantly renders military hardware obsolete, and horror stories convince most of the public that continued escalation of the arms race is absolutely necessary for survival.
3. This does not increase the productive capacity of the capitalist economy.
4. Because virtually all military production is done by giant private corporations or subcontracted to smaller private corporations, it does not compete with private profit.
5. Although precise statistics to substantiate this are unobtainable, most evidence points to military production being more profitable than production for the free market.
6. As Rosa Luxemburg pointed out, large military establishments are necessary to maintain and enlarge the worldwide "sphere of influence" of capitalist countries, in which profitable foreign investment and very favorable terms of trade can be assured.
7. As Veblen pointed out, jingoism, patriotism, and militarism are perhaps the most effective means of keeping workers docile and promoting the view among workers that their interests are in harmony with the interests of the capitalists.

### The Debt Economy

The post-World War II improvement in the American economy is largely attributed to increased debt expansion, as demonstrated by Keynes. He highlighted how demand constraints, could hinder an economy's output. Keynesian fiscal policy, advocates for government expenditure expansion financed through borrowing to stimulate demand. However, heavy reliance on debt, whether public or private, to fuel demand has led to adverse effects on economies, despite initially boosting utilization of resources.

Following World War II, the United States government faced significant debt, reaching 109 percent of GDP in 1946. However, from 1946 to 1974, the debt relative to GDP consistently decreased, largely due to robust economic growth, averaging around 7 percent annually. The trend reversed in the 1980s, particularly during Reagan's presidency, with federal debt relative to GDP rising from 26 percent in 1980 to 41 percent by the end of his term. This period of increased debt expansion coincided with economic recovery, akin to the post-World War II era, providing support for Keynesian fiscal policies.

Federal debt relative to GDP continued to rise until 1993. President Clinton implemented measures to reduce the debt, aiming to lower long-term interest rates and stimulate the economy. The debt-to-GDP ratio stabilized around 49 percent from 1993 to 1995 before declining significantly, reaching 32.5 percent in 2001. The decrease in the debt ratio was not

solely due to GDP growth but also to a consistent reduction in the annual growth rate of federal debt, turning negative by 1998.

After 1992, the economy expanded and government debt decreased, although overall debt-to-GDP ratio stayed mostly stable. While federal debt growth decreased, total debt (including households, businesses, and government) relative to GDP remained stable, starting at 186 percent in 1992 and ending the decade at 185 percent. Additionally, there was a notable increase in the annual growth rate of business debt during this period, indicating another sector contributing to the maintenance of total debt levels.

From the post-World War II era until the mid-1960s, U.S. households experienced a steady increase in debt relative to disposable income, reaching 68 percent in 1965. This trend flattened until the mid-1980s but then rose again, with household debt reaching 94 percent of disposable income by the end of the 1990s. Notably, this increase in household debt was primarily driven by consumer credit rather than mortgage debt. Consumer credit as a proportion of disposable income rose from around 17 percent in 1992 to about 23 percent by the end of the 1990s. Thus, we can say that the decline of federal debt during 1990s did not refute the Keynesian analysis, it merely represented a change in the composition of U.S. debt which was needed to fuel the economy.

Keynes advocated for fiscal conservatism, emphasizing the importance of balancing government deficits during downturns with surpluses during upturns. He envisioned impartial government bureaucrats making economic policy decisions, but political reluctance to implement austerity measures often prevents long-term fiscal balance. Furthermore, within capitalism the government tends to be influenced by wealthy individuals and powerful corporations. Additionally, government debt can lead to market distrust, rising interest rates, and fears of default, as seen in other countries beyond the United States.

Keynes's analysis overlooked the longer-term implications of businesses responding to demand constraints. He did not explore how businesses foster consumerism and expand consumer credit to finance it, leading to a pyramid of credit within lending institutions. This oversight left Keynes unable to anticipate the risks associated with the growth of private debt, including the possibility of a significant default that may set off a worldwide chain reaction that would cause a horrible financial crisis.

Household debt in the US increased from 2000 to 2007, with an average annual growth rate of 10%, reaching nearly 133% of disposable income by 2007. This increase was primarily driven by rising mortgage debt, which went up to over 101% of disposable income from about 65% during the same period. Businesses, especially banks, devised new methods to finance mortgage debt, leading to substantial profits until the financial crisis struck.

Following the financial crisis, household debt growth halted and turned negative by 2009 as credit markets froze. Central banks responded with Keynesian measures, injecting money and lowering interest rates. Public debt went up as private debt stagnated, driven mainly by the

federal government's actions such as tax cuts and bailouts. Federal debt relative to GDP rose from 32.5% in 2001 to 53% by 2009, growing over 20% annually during 2008-2009. During this period, both fiscal and monetary measures were taken to prevent a banking system collapse to prevent a depression.